

difference, however, is that rather than employing human market makers the AMM system uses advanced computer systems to enter quotes and facilitate trades. The registered AMM still has an obligation to maintain a fair and orderly market, provide liquidity when needed, and provide market quotes a specified percentage of the time.

AMM systems automatically enter bids and offers into the market. After the AMM system transacts with a market participant they become either long or short shares and they will seek to offset any acquired position through further usage of limit orders. The AMM systems look to profit from buying at the bid and selling at the offer and earning the full spread. And as an incremental incentive, registered auto market maker firms are often provided an incremental rebate for providing liquidity. Therefore, they can profit on the spread plus rebates provided by the exchange. This is also causing some difficulty for portfolio managers seeking to navigate the price discovery process and determine fair value market prices.

AMM black box trading models will also include an alpha model to help forecast short-term price movement to assist them in determining the optimal holding period before they are forced to liquidate an acquired position to avoid a loss. For example, suppose the bid-ask spread is \$30.00 to \$30.05 and the AMM system bought 10,000 shares of stock RLK at the bid price of \$30.00. If the alpha forecast expects prices to rise the AMM will offer the shares at the ask price of \$30.05 or possibly higher in order to earn the full spread of \$0.05/share or possibly more. However, if the alpha forecast expects prices to fall, the AMM system may offer the shares at a lower price such as \$30.04 to move to the top of the queue or if the signal is very strong the AMM systems may cross the spread and sell the shares at \$30.00 and thus not earn a profit, but not incur a loss either.

Most AMM traders prefer to net out all their positions by the end of the day so that they do not hold any overnight risk. But they are not under any obligation to do so. They may keep positions open (overnight) if they are properly managing the overall risk of their book or if they anticipate future offsetting trades/orders (e.g., they will maintain an inventory of stock for future trading). Traditional AMM participants continue to be concerned about transacting with an informed investor, as always, but it has become more problematic with electronic trading since it is more difficult to infer if the other side is informed (has strong alpha or directional view) or uninformed (e.g., they could be a passive indexer required to hold those number of shares), since the counterparty's identity is unknown.

Some of the main differences between AMM and traditional MM are that AMM maintains a much smaller inventory position, executes smaller